

We chose Microservices Architecture for the Vaccine Registration and Booking System.

- **Architectural Choice:** Microservices Architecture.
- **Two Reasons:**
 1. **Independent Scalability for Traffic Spikes:** The system must handle massive, sudden loads of up to 10,000 concurrent booking requests during national rollout periods (REQ-011). Microservices allow us to scale just the "Scheduling & Booking Service" instances to handle the traffic surge without needing to waste resources scaling the entire system.
 2. **Fault Isolation for High Availability:** To maintain the strict 99.5% uptime SLA (REQ-012), this architecture ensures that a failure in one component does not bring down the whole application. For example, if the third-party SMS gateway used by the "Notification Service" crashes, citizens can still register and book appointments without interruption.
- **Security Advantage:** This architecture natively supports strict data segmentation, directly addressing the requirement to protect sensitive citizen health and identity data (REQ-013). By placing the "Identity & Registration Service" and the "Immunization Record Service" in their own isolated layers with dedicated databases, sensitive Personal Identifiable Information (PII) is kept separate from the public-facing booking logic. Even if the scheduling endpoint is compromised, the attacker does not gain direct access to the encrypted clinical records.
- **Performance Benefit:** Microservices enable asynchronous, event-driven processing which guarantees fast user response times. For example, when a user confirms a booking, the Scheduling Service can instantly return a success message to the UI (meeting the <2s response time in REQ-011) and simply send an event to the Notification Service to trigger the email/SMS reminder (REQ-005) in the background. The user never has to wait on the screen for the email server to respond